

Assignment 1: Create a Spark Cluster on Google Cloud

Evaluation: 10% of the grade

In this lab assignment you will learn how to create a Dataproc cluster, run a basic [Apache Spark](#) job in the cluster, and then modify the number of workers in the cluster, using Google Cloud console and Google Cloud SDK also known as gcloud CLI.

Tasks to perform:

Step 1: Create a Dataproc cluster

Step 2: Add and submit a simple Spark Job

Step 3: Verify job output

Step 4: Update the cluster

Don't forget to clean up the resources

Repeat steps 1-4 using GC console as well as gcloud CLI following instruction at links below:

https://cloud.google.com/dataproc/docs/quickstarts/create-cluster-console#update_a_cluster

<https://cloud.google.com/dataproc/docs/quickstarts/create-cluster-gcloud>

Submit your report along with the supporting material by February 19, 2025.

Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Workflow	Followed all steps in the work flow	Followed most steps in the workflow	No order in the workflow	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	7
Report	Well organized and easy to follow report with meaningful section headings	Somewhat organized, a little hard to follow and fair section headings	Not organized, very hard to follow and messy report	2
Total				10